

Venkata Bhupathi

Lead Software Engineer | Cloud, DevOps & Identity | .NET and Azure

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SUMMARY

Full-stack engineer with 10+ years building and running production software in C#/.NET, Azure, and Kubernetes. Technical lead for Kumo, an enterprise cloud-storage platform used by 18 universities and about 400,000 students, and lead engineer for Go.IU, Indiana University's URL-shortening service. Own the full lifecycle: architecture, delivery pipelines, identity federation, server and certificate operations, and partner-facing work such as billing and contract renewals.

EXPERIENCE

Full-Stack Software Engineer

Feb 2020 – Present

Indiana University, Indianapolis, IN (with the university since Aug 2018)

- **Lead Kumo (IU CloudStorage)**, a cross-platform platform that mounts Box, Google Drive, Dropbox, OneDrive, OneDrive for Business, and FileShare as network drives on managed lab machines, tied to campus sign-in. Serves 18 universities and about 400,000 students; technical lead since 2018.
- **Modernizing the platform:** moving .NET Framework code to .NET Core and Blazor, rewriting the Kumo Portal on a modern stack, and automating deployment. Led the HMAC authentication migration and the CBFS file-system driver migration, and led the ADFS versus Microsoft Entra evaluation.
- **Led Go.IU (go.iu.edu) end to end:** IU's URL shortener. Affiliates sign in to create links and anyone can follow them. Runs in Docker on Kubernetes, deployed with Helm, on SQL Server.
- **Contributor on university applications:** Unified Copy Machine & Print (inventory and invoicing; Docker, Kubernetes, Helm), RivetBlazor (Blazor components for IU's Rivet design system), and IT People (IT staff directory).
- **Operations and security:** manage 12+ servers and the ADFS environment, including the SSL/TLS certificate lifecycle and renewals. Configure SAML, WS-Federation, and ADFS federation with external institutions using PowerShell.
- **Partner management:** own billing and MSA renewals for external universities that use Kumo, and work directly with their teams to clarify requirements and resolve complex issues.
- **Engineering practice:** Azure DevOps YAML pipelines for build, test, and deploy; Docker and Helm packaging; Azure Functions, including a serverless HTTP API in Python; xUnit and NUnit tests; code review; root-cause analysis across application, database, authentication, and infrastructure layers.

Web & Software Developer

Aug 2018 – Jan 2020

Indiana University, Indianapolis, IN

- Built and supported the Kumo cross-platform storage connector using .NET, C#, JavaScript, Electron, SQL Server, and virtual file-system technology across Windows, macOS, and Linux.
- Integrated Box, Google Drive, Dropbox, and OneDrive; configured ADFS federation with external institutions using PowerShell; worked with Azure Service Bus and SQL Server; resolved production issues for client campuses.

Earlier .NET roles

2012 – 2018

Software Developer, Control Station, Inc., Manchester, CT (Apr – Jul 2018). WPF/MVVM and ASP.NET MVC applications with Entity Framework and NUnit.

Application Developer, Eniac Systems, Inc., Edison, NJ (Dec 2017 – Mar 2018). ASP.NET MVC and WCF REST applications; tuned SQL Server procedures and triggers.

Software Engineer, GP-Technologies LLC, Columbia, SC (Apr – Nov 2017). C# business and data-access components, REST services, and React UI.

Software Developer, Blueliner Software Services, Hyderabad, India (Sep 2012 – Jul 2015). ASP.NET MVC, AngularJS, and ReactJS applications on SQL Server and IIS across the full SDLC.

CERTIFICATIONS

Microsoft Certified: DevOps Engineer Expert (AZ-400) and Azure Developer Associate (AZ-204)

Scrum Alliance: Certified ScrumMaster® (CSM®)

Indiana University: GenAI 101 Certified (2025)

EDUCATION

M.S., Computer Information Technology , Central Connecticut State University, New Britain, CT	2016
B.Tech, Information Technology , Jawaharlal Nehru Technological University, Kakinada, India	2012

SKILLS

Languages and frameworks: C#, .NET and .NET Core, ASP.NET Core and MVC, Blazor, React, JavaScript, Python, SQL, Entity Framework, REST and Web API

Cloud and delivery: Azure, Azure DevOps (YAML CI/CD), Azure Functions, Azure Service Bus, Docker, Kubernetes, Helm, PowerShell

Identity and systems: ADFS, SAML, WS-Federation, Microsoft Entra, Active Directory, HMAC request signing, IIS, Windows Server, Linux

Quality and practice: xUnit, NUnit, unit and integration testing, Agile, Scrum, code review, production support

AFFILIATIONS

Fellow, Hackathon Raptors (Community Interest Company), 2024: recognized for contributions to cloud computing and multi-institution cloud infrastructure systems.